When Intelligence Meets Power: Governing the Co-Evolution of Artificial Intelligence and Post-Quantum Cryptography

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1. Introduction and Motivation

Artificial intelligence (AI) and quantum computing are transforming how systems learn, compute, and secure information. While AI continues to enhance reasoning, prediction, and decision-making, quantum computing also introduces new levels of power that could disrupt the cryptographic methods protecting these same systems. This tension between intelligence and computational strength defines the future of secure computing.

My research examines this interaction through the study of post-quantum cryptography and artificial intelligence governance. The goal is to explore how artificial intelligence and quantum-resistant cryptography influence one another. The growing computational power of AI could accelerate cryptanalysis and expose vulnerabilities faster than humans can evaluate them. At the same time, it can be used to strengthen cryptographic resilience through automated key management, anomaly detection, and adaptive defense.

This tension underscores the need for governance frameworks that promote safe progress without stifling innovation, ensuring that intelligence and encryption evolve together under shared principles of privacy, accountability, and trust.

2. Research Problem and Objectives

Artificial intelligence and post-quantum cryptography are advancing rapidly yet often in isolation. As AI becomes more computationally capable, it could exploit weaknesses in cryptographic systems as easily as it could help defend them. This duality presents a paradox: as we build cryptography to protect AI, we are also creating AI systems with the capacity to challenge that protection. Effective governance must therefore balance innovation and accountability, ensuring that both technologies evolve together responsibly.

The core goal of this research is to explore how artificial intelligence and post-quantum cryptography can evolve together within secure and ethical governance frameworks. Rather than treating them as separate frontiers, this study approaches them as interdependent technologies that demand coordinated design and oversight.

2.1 Research Question

How can artificial intelligence and post-quantum cryptography evolve together under governance frameworks that preserve privacy, transparency, and innovation?

2.2 Objectives

- Examine how artificial intelligence can accelerate or mitigate cryptanalysis in both classical and quantum-resistant environments.
- Evaluate the computational feasibility of integrating post-quantum algorithms such as CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+ within AI model training, inference, and communication pipelines
- Explore how artificial intelligence methods can enhance cryptographic security through automated key management, anomaly detection, and adaptive threat modeling
- Assess existing policies and international regulations governing AI and cryptography to identify gaps that hinder quantum-era preparedness.
- Develop a governance framework that aligns technological progress with ethical, social, and legal safeguards.

3. Methodology

This project combines technical experimentation with policy and governance analysis to study the interaction between artificial intelligence and post-quantum cryptography. The goal is to understand how these systems perform together at the algorithmic level as well as how their development can be guided by effective and forward-thinking governance

3.1 Technical Component

The technical study will integrate post-quantum cryptographic algorithms (CRYSTALS-Kyber, CRYSTALS-Dilithium, SPHINCS+) into AI systems using standard quantum and ML libraries such as Qiskit, LibOQS, and PyTorch. Simulations will measure latency, accuracy, and resilience during secure model communication and federated learning. Machine-learning models will also be trained to detect weaknesses in quantum-resistant schemes, revealing how AI can both accelerate and mitigate cryptanalysis.

3.2 Policy and Governance Component

The policy analysis will review global efforts, including the National Institute of Standards and Technology (NIST) post-quantum cryptography initiative and emerging AI regulations such as the EU AI Act. The study will identify regulatory gaps around accountability, export control, and data security where current frameworks fail to anticipate quantum-era risks. Insights from organizations like the OECD and World Economic Forum will inform a governance model that aligns technical resilience with ethical oversight.

3.3 Integration and Framework Design

Findings from the technical and policy components will inform a *Governance Framework for Secure Co-Evolution* that connects technical resilience with ethical and regulatory oversight. The framework will guide policymakers, engineers, and researchers in developing standards that enable responsible AI and post-quantum cryptography evolution.

4. Anticipated Outputs

This research will produce both technical and policy-oriented deliverables that can demonstrate how artificial intelligence and post-quantum cryptography can develop together under shared governance principles. The aim is to contribute tools, frameworks, and insights that are practical for engineers while informative for policy makers.

4.1 Technical Deliverables

- A prototype integrating post-quantum cryptographic algorithms into AI systems, demonstrating secure communication, encryption, and authenticated exchanges in federated learning environments.
- Experimental datasets and performance reports that compare latency, accuracy, and computational costs of AI systems secured with post-quantum methods versus classical encryption.
- An AI tool trained to detect weaknesses in post-quantum algorithms and evaluate how machine learning accelerates cryptanalysis.

4.2 Policy and Governance Deliverables

- A white paper on the *Governance Framework for Secure Co-Evolution*, recommending alignment of AI and cryptographic development with international standards and ethical safeguards.
- A policy brief summarizing the most pressing regulatory gaps and proposing mechanisms for cooperation among standards bodies such as NIST, ISO, and the OECD.
- A public-facing summary designed for interdisciplinary audiences explaining the societal relevance of responsible co-development between intelligence and encryption technologies.

4.3 Academic and Community Impact

- A research article submission to a journal or conference in cybersecurity or technology policy (e.g. IEEE Security & Privacy or the Journal of Cyber Policy).
- A presentation or workshop within the Berkman Klein community to foster discussion between technologists, ethicists, and legal scholars on governing quantum-ready AI systems.

5. Relevance to the Berkman Klein Center

This project directly aligns with the Berkman Klein Center's focus on emerging technologies, ethics, and governance. The intersection of artificial intelligence and post-quantum cryptography engages several of the Center's research priorities, including AI governance, technology and global security, and responsible innovation. By bridging technical and policy dimensions, this project supports BKC's mission to shape technologies that reinforce public trust and serve the public interest.